EMG VEST / ESP32 + 28-CHANNEL ACQUISITION

Exact GPIO nets and official DevKitC V4 header positions. For 30-pin clones, follow GPIO labels, not these header positions.

U0 ESP32-WROOM-32 / -32E

ESP32-DevKitC V4

Built-in Bluetooth + Wi-Fi

No P4 or C6 coprocessor required

U1-U4 ADS1299 acquisition board

Four 8-channel ADCs; use 28 of 32 inputs

SPI Mode 1 | 3.3 V logic | 2 kSPS/channel

GPIO18 / J3-9	SCLK	ALL: pin 40
GPIO23 / J3-2	MOSI	ALL: DIN pin 34
GPIO19 / J3-8	MISO	ALL: DOUT pin 43
GPIO25 / J2-9	CS0_N	U1: CS pin 39
GPIO26 / J2-10	CS1_N	U2: CS pin 39
GPIO27 / J2-11	CS2_N	U3: CS pin 39
GPIO32 / J2-7	CS3_N	U4: CS pin 39
GPIO34 / J2-5	DRDY_N	U1 ONLY: pin 47
GPIO33 / J2-8	START	ALL: pin 38
GPIO13 / J2-15	RESET_N	ALL: pin 36
GND / J2-14	GND	ALL: DGND / AVSS

CLOCK + CONTROL

One 2.048 MHz / 3.3 V oscillator:

OUT -> U1/U2/U3/U4 CLK (pin 37).

100 nF bypass at oscillator supply.

Each CLKSEL (52) -> GND through 10k.

Each PWDN_N (35) -> 3.3V_D through 10k.

Each CS_N -> 3.3V_D through 10k.

RESET_N -> 3.3V_D through 10k.

START -> GND through 10k.

DRDY_N -> 3.3V_D through 10k.

CHANNEL BANKS

U1 / CS0: IN1-IN8 -> CH01-CH08

U2 / CS1: IN1-IN8 -> CH09-CH16

U3 / CS2: IN1-IN8 -> CH17-CH24

U4 / CS3: IN1-IN4 -> CH25-CH28

U4 IN5-IN8 are unused: disable channels;

terminate unused analog inputs per TI.

Only U1 DRDY is wired to the ESP32.

Never short four DRDY outputs together.

IMPLEMENTATION NOTES

GPIO34 is input-only: external pull-up required.

Keep all four ADCs on one compact PCB.

Common clock + START synchronize conversion.

Shared DOUT requires tri-state when CS is high.

Do not use GPIO6-11: connected to flash.

Avoid boot strap pins for these controls.

3.3 V GPIO only; never feed 5 V into a GPIO.

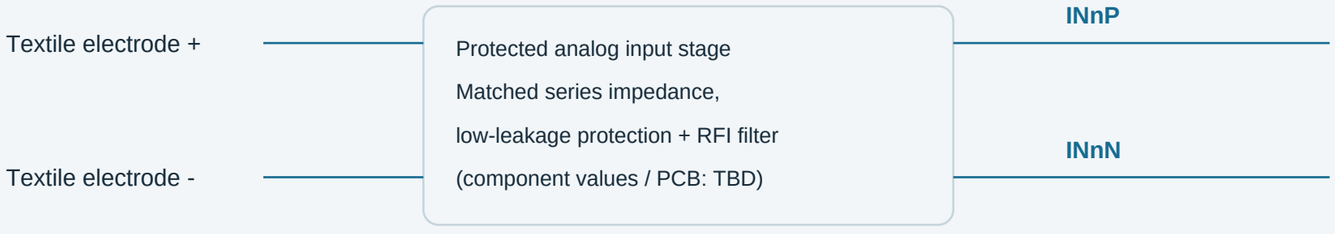
P4 firmware must be ported to this pin map.

Use BLE UART for both Android and iPhone.

EMG VEST / ELECTRODES + MUSCLE CHANNELS

28 independent differential measurements = 56 measurement contacts, plus the separately engineered bias/reference contact.

REPEAT THIS INPUT PAIR FOR EACH CHANNEL



Measurement contacts go to the AFE, never directly to ESP32 GPIO or board ground.

BIAS / REFERENCE IS A SEPARATE CIRCUIT

Body bias contact -> protected bias circuit.

Unipolar AFE requires a controlled common mode.

Draft assumes an external mid-supply bias stage;

ADS1299 internal BIAS amplifier remains OFF.

Do not join every negative electrode to REF.

Bias, input protection and leakage need validation.

ID	Side	Muscle	ADC input	P / N pin
CH01	L	Anterior deltoid	U1 IN1	16 / 15
CH02	R	Anterior deltoid	U1 IN2	14 / 13
CH03	L	Middle deltoid	U1 IN3	12 / 11
CH04	R	Middle deltoid	U1 IN4	10 / 9
CH05	L	Posterior deltoid	U1 IN5	8 / 7
CH06	R	Posterior deltoid	U1 IN6	6 / 5
CH07	L	Biceps brachii	U1 IN7	4 / 3
CH08	R	Biceps brachii	U1 IN8	2 / 1
CH09	L	Triceps brachii (long head)	U2 IN1	16 / 15
CH10	R	Triceps brachii (long head)	U2 IN2	14 / 13
CH11	L	Pectoralis major (clavicular)	U2 IN3	12 / 11
CH12	R	Pectoralis major (clavicular)	U2 IN4	10 / 9
CH13	L	Upper trapezius	U2 IN5	8 / 7
CH14	R	Upper trapezius	U2 IN6	6 / 5

ID	Side	Muscle	ADC input	P / N pin
CH15	L	Middle trapezius	U2 IN7	4 / 3
CH16	R	Middle trapezius	U2 IN8	2 / 1
CH17	L	Lower trapezius	U3 IN1	16 / 15
CH18	R	Lower trapezius	U3 IN2	14 / 13
CH19	L	Latissimus dorsi	U3 IN3	12 / 11
CH20	R	Latissimus dorsi	U3 IN4	10 / 9
CH21	L	Serratus anterior	U3 IN5	8 / 7
CH22	R	Serratus anterior	U3 IN6	6 / 5
CH23	L	Rectus abdominis	U3 IN7	4 / 3
CH24	R	Rectus abdominis	U3 IN8	2 / 1
CH25	L	External oblique	U4 IN1	16 / 15
CH26	R	External oblique	U4 IN2	14 / 13
CH27	L	Lumbar erector spinae	U4 IN3	12 / 11
CH28	R	Lumbar erector spinae	U4 IN4	10 / 9

P / N pin numbers above refer to ADS1299 TQFP-64. Left/right are the wearer's anatomical sides.

Keep each electrode pair together in the harness. Make electronics detachable for washing. Keep the antenna away from conductive fabric.

EMG VEST / POWER + OPTIONAL IMUs

Battery-only acquisition while electrodes are worn. Disconnect USB, charging cables and grounded bench instruments.

Protected 1S battery

Battery + fuse + power switch

Power conversion module

Regulated 5V_SYS for DevKitC

Quiet 5V_A + regulated 3.3V_D

POWER NETS

5V_SYS -> DevKitC J2-19 (5V); GND -> J2-14

5V_A -> AFE AVDD; 3.3V_D -> AFE DVDD / IMUs

All electronic grounds join; never parallel regulator outputs.

Do not power the DevKitC from USB and its 5V header at the same time. The two regulated 5 V nets may require separate filtering.

ADS1299 SUPPORT CONNECTIONS - EACH CHIP

AVDD: 19,21,22,56,59; AVDD1: 54 -> 5V_A.

AVSS: 20,23,32,57,58; AVSS1: 53 -> GND.

DVDD: 48,50 -> 3.3V_D; DGND: 33,49,51 -> GND.

VREFN 25 -> GND; >=10uF from VREFP 24 to VREFN.

VCAP1 28: 100uF; VCAP2 30: 1uF; VCAP4 26: 1uF to AVSS.

VCAP3 55: 1uF || 100nF to AVSS; keep capacitors close.

Bypass supply groups locally: 1uF || 100nF, per TI layout.

DAISY_IN 41 and RESV1 31 -> DGND; SRB1/SRB2 disabled.

NC 27,29 and reserved 64: leave open. GPIO1-4: 10k to GND.

OPTIONAL 3 x SIX-AXIS IMU - I2C THROUGH A MULTIPLEXER

ESP32 GPIO21 / J3-6 -> TCA9548A SDA (TSSOP pin 23).

ESP32 GPIO22 / J3-3 -> TCA9548A SCL (TSSOP pin 22).

TCA VCC24 -> 3.3V_D; GND12 -> GND; A0/A1/A2 -> GND.

RESET3 -> 3.3V_D via 10k; address = 0x70.

SD0/SC0 (4/5) -> chest LSM6DSOX SDA/SCL.

SD1/SC1 (6/7) -> left upper-arm LSM6DSOX SDA/SCL.

SD2/SC2 (8/9) -> right upper-arm LSM6DSOX SDA/SCL.

Each LSM6DSOX: SDA14, SCL13, VDD8/VDDIO5 -> 3.3V_D.

CS12 -> 3.3V_D; SA0 pin1 -> GND (0x6A); GND6/7 -> GND.

POWER / BODY-INTERFACE DESIGN STATUS

This drawing defines interconnects, not a production PCB.

Power converter parts, analog input protection and bias-stage

component values still need selection and electrical verification.

Do not connect unprotected bare ADS1299 inputs to the body.

Internal BIASOUT pins are not tied together or wired to the body.

IMU BUS DETAILS

Use 4.7k pull-ups to 3.3V_D on SDA/SCL upstream and on each

used branch (account for any pull-ups already on breakout boards).

Add 100nF at each supply. Select one mux branch at a time.

Use short wires; validate rise time before running at 400 kHz.

IMU acquisition is optional and needs additional firmware.

FIRMWARE PIN ASSIGNMENT

SCK=18 | MOSI=23 | MISO=19 | CS={25,26,27,32} | DRDY=34 | START=33 | RESET=13

I2C SDA=21 | I2C SCL=22 | use VSPI / SPI Mode 1 for ADS1299; all four chips share the external clock.

VERIFIED PRIMARY REFERENCES

ESP32 header pinout

TCA9548A datasheet

ADS1299 pinout / support circuit

LSM6DSOX datasheet